

Likelihood of the Evidence given that the Hypothesis is True

Prior Probability of the Hypothesis

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$

Posterior Probability of the Hypothesis given that the Evidence is True

Prior Probability that the evidence is True

f vector $x = (x_1, x_2, \dots, x_n)$

y

$$P(y|x) = \frac{P(x|y) * P(y)}{P(x)}$$

0
1

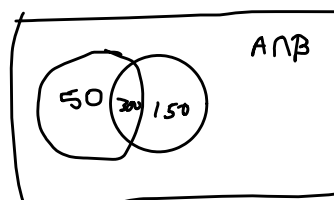
$$P(y|x) = \frac{P(x_1|y) * P(x_2|y) \dots P(x_n|y) * P(y)}{P(x_1) * P(x_2) \dots P(x_n)}$$

$$P(y|x_1, x_2, \dots, x_n) = P(y) \prod_{i=1}^n P(x_i|y)$$

$$y = \arg \max_y P(y) \prod P(x_i|y)$$

	Actually purchased		total
	Y	N	
Planned to purchase			
Y	150	200	350
N	200	750	950
total	350	950	1300

y
450



$$P(A) = \frac{350}{1300}$$

$$P(B) = \frac{450}{1100}$$

$$P(A \cap B) = \frac{260}{1100}$$

$$P(B|A)$$

$$P(A \cap B) = P(B|A) \cdot P(A)$$

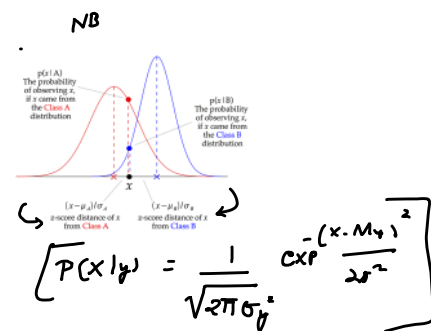
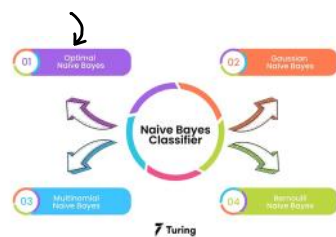
LIKELIHOOD
The probability of "B" being True, given "A" is True

PRIOR
The probability "A" being True. This is the knowledge.

POSTERIOR
The probability of "A" being True, given "B" is True

MARGINALIZATION
The probability "B" being True

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$



K-Nearest Neighbors



1. K set

2.
3.



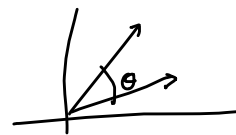
$$\hookrightarrow \text{Knn} \rightarrow d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

$$d(p, q) = \sum_{i=1}^n |p_i - q_i|$$

$$\text{Minkowski} : d(p, q) = \left(\sum_{i=1}^n |p_i - q_i|^r \right)^{1/r}$$

Cosine $\text{sim}(p, q) = \frac{p \cdot q}{\|p\| \|q\|}$

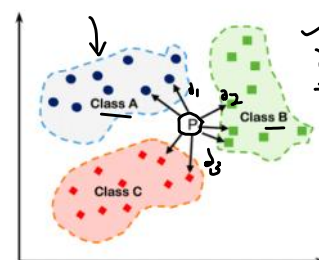
Text



→ setosa
↙
↘
versicol

→ class similarity
→ fare map

K Nearest Neighbors



$$d_1 < d_2 < d_3$$

